$$\frac{\sin\phi_1}{\sin\phi_2} = \frac{n_2}{n_1} \qquad (4.1)$$

where n_1 and n_2 are the refractive indices of the two media (see Fig. 4.1). Although there is no additional physics involved (other than the Snell's laws) in the tracing of rays, the design of even a simple optical system involves tracing many rays and therefore considerable numerical computations. Nowadays, such numerical computations are usually done on a high-speed computer. It may be of interest to note that optical designers were among the first to make use of electronic computers when they were introduced in the early 1950s.

4.2 REFRACTION AT A SINGLE SPHERICAL SURFACE LO 1

We will first consider refraction at a spherical surface SPM separating two media of refractive indices n_1 and n_2 (see Fig. 4.1(a)). Let C represent the center of curvature of the spherical surface. We will consider a point object O emitting rays in all directions. We will use Snell's laws of refraction to determine the image of the point O. We may mention that not all rays emanating from O converge to a single point; however, if we consider only those rays which make small angles with the line joining the points O and C then all rays do converge to a single point I [see Fig. 4.1(a)]. This is known as the *paraxial approximation* and according to Fermat's principle, all paraxial rays take the *same* amount of time to travel from O to I (see Example 3.3).

Now, in terms of the angles defined in Fig. 4.1(a) we have

$$\phi_1 = \beta + \alpha_1 \quad \text{and} \quad \phi_2 = \beta - \alpha_2$$

We next make use of the paraxial approximation, viz., all angles ϕ_1, ϕ_2, α_1, α_2 and β are small, so that we may write

$$\sin\phi_1 \approx \tan\phi_1 \approx \phi_1 \text{ etc.}$$

where the angles are obviously measured in radians. Thus, we have

$$\sin\phi_1 \approx \phi_1 = \beta + \alpha_1 \approx \tan\beta + \tan\alpha_1 \approx \frac{h}{r} + \frac{h}{x} \qquad (4.2)$$

and

$$\sin\phi_2 \approx \phi_2 = \beta - \alpha_2 \approx \tan\beta - \tan\alpha_2 \approx \frac{h}{r} - \frac{h}{y} \qquad (4.3)$$

where the distances h, x, y and r are defined in Fig. 4.1(a) and we have assumed that the foot of the perpendicular (D) is very close to the point P so that $OD \approx OP = x$, $ID \approx IP = y$, etc. We now use Eqs. (4.1) – (4.3) to obtain (in the paraxial approximation):

$$n_1\left(\frac{h}{r} + \frac{h}{x}\right) = n_2\left(\frac{h}{r} - \frac{h}{y}\right)$$

or

$$\frac{n_2}{y} + \frac{n_1}{x} = \frac{n_2 - n_1}{r} \qquad (4.4)$$

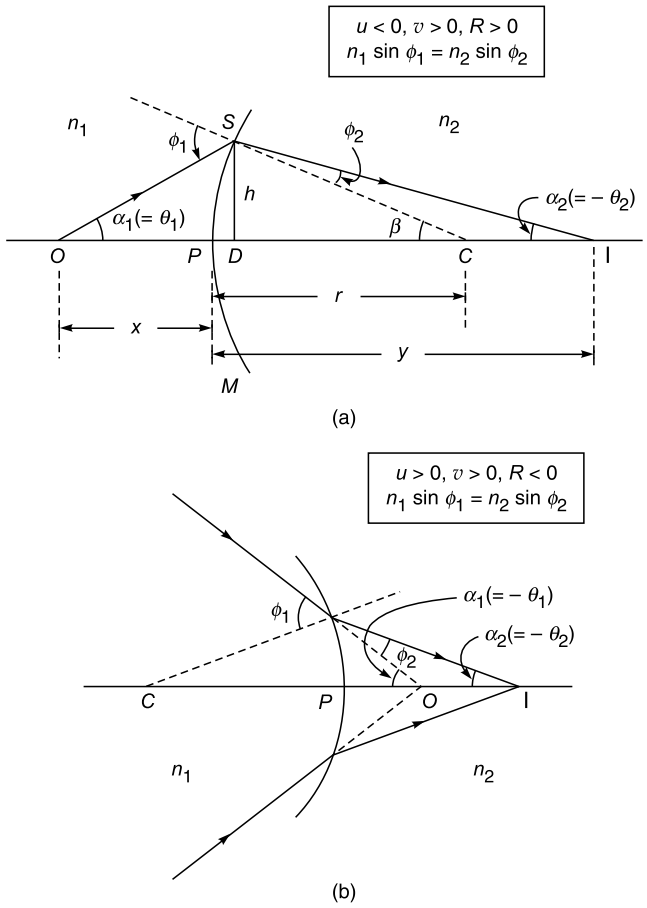


Fig. 4.1 (a) Paraxial image formation by a spherical refracting surface separating media of refractive indices n_1 and n_2. O represents the object point and I the paraxial image point. (b) corresponds to positive u.

4.2.1 The Sign Convention

Before we proceed further, we should state the sign convention which we will be using throughout in the book. We refer to Fig. 4.1(a) and consider the point P as the origin of the co-ordinate system. The sign convention is as follows:

1. The rays are always incident from the left on the refracting (or reflecting) surface.
2. All distances to the right of the point P are positive and distances to the left of the point P are negative. Thus in Fig. 4.1 (a), the object distance u is a negative

quantity and the image distance v and the radius of curvature R are positive quantities. For u to be positive, we must have a situation like the one shown in Fig. 4.1(b); in the absence of a refracting surface, the rays converge to a point to the *right* of P.

3. The angle that the ray makes with the axis is positive if the axis has to be rotated in the anticlockwise direction (through the acute angle) to coincide with the ray. Conversely, if the axis has to be rotated in the clockwise direction (through the acute angle) to coincide with the ray, then the slope angle is negative. Thus in Fig. 4.1(a), if θ_1 and θ_2 are the angles that the rays OS and SI make with the axis, then $\theta_1 = \alpha_1$ and $\theta_2 = -\alpha_2$; α_1, α_2 and β represent the magnitudes of the angles. (If the final result does not depend on the angles, then it is more convenient to use the magnitude of the angles as has indeed been done in Sec. 4.2). In Fig. 4.1(b), both θ_1 and θ_2 are negative quantities.
4. The angle that a ray makes with the normal to the surface is positive if the normal has to be rotated in the anticlockwise direction (through the acute angle) to coincide with the ray, and conversely. Thus, in Fig. 4.1 (a), ϕ_1 and ϕ_2 are positive quantities.
5. All distances measured upward from the axis (along a perpendicular to the axis) are positive, and all distances measured in the downward direction are negative.

4.2.2 The Gaussian Formula for a Single Spherical Surface

If we now use the sign convention discussed above, then for the ray diagram shown in Fig. 4.1(a), $u = -x$, $v = y$ and $R = r$. Thus Eq. (4.4) becomes

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \qquad \textbf{(4.5)}$$

which gives the image point due to refraction at a spherical surface (see also Eq. 3.10 of the previous chapter). Equation (4.5) is known as the **Gaussian formula for a single spherical surface.** It should be noted that, corresponding to Fig. 4.1(a), u is negative and v positive, whereas for Fig. 4.1(b) u and v are both positive.

Example 4.1 Consider a medium of refractive index 1.5 bounded by two spherical surfaces $S_1P_1M_1$ and $S_2P_2M_2$ as shown in Fig. 4.2. The radii of curvature of the two surfaces are 15 and 25 cm with their centers at C_1 and C_2, respectively. There is an object at a distance of 40 cm (from P_1) on the line joining C_1 and C_2. Determine the position of the paraxial image.

Solution: We first consider refraction by $S_1P_1M_1$. Obviously $u = -40$ cm, $R = +15$ cm, $n_1 = 1.0$ and $n_2 = 1.5$. Thus,

$$\frac{1.5}{v} + \frac{1}{40} = \frac{0.5}{15} \quad \Rightarrow v = +180 \text{ cm}$$

In the absence of the second surface, the image is formed at O' at a distance of 150 cm from P_2. O' now acts as a virtual object and since it is to the right of $S_2P_2M_2$ we have, while considering refraction by the second surface, $u = +150$ cm, $R = -25$ cm, $n_1 = 1.5$ and $n_2 = 1.0$. Thus,

$$\frac{1.0}{v} - \frac{1.5}{150} = +\frac{0.5}{25}$$

giving

$$v = +33\tfrac{1}{3} \text{ cm}$$

and a real image is formed on the right of P_2 at a distance of $33\frac{1}{3}$ cm.

We should point out that while considering refraction by a single surface (as in Fig. 4.1), the axis of the system is defined by the line joining the object point O and the center of curvature C; thus *any* ray from the point O (like OS) will be in the plane containing the axis and the normal at the point S and consequently, the refracted ray will *always* intersect the axis. On the other hand, if there is second refracting surface (like in Fig. 4.2 or as in a lens) then the line joining the two

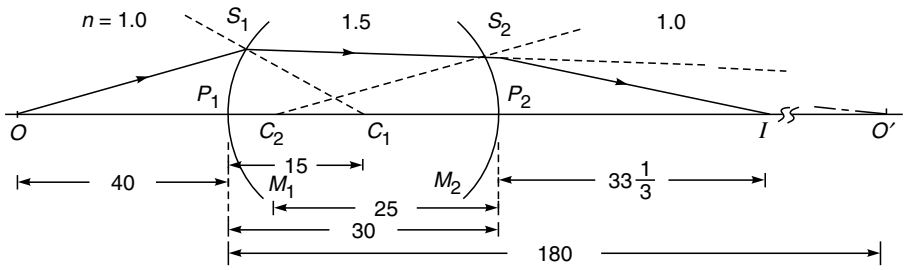


Fig. 4.2 Paraxial image formation by a medium of refractive index 1.5 bounded by two spherical surfaces $S_1P_1M_1$ and $S_2P_2M_2$. All distances are measured in centimeters.

centers of curvature is defined as the axis. In the latter case, as can be readily seen, not all rays from an off-axis point will intersect the axis and after refraction at the second surface will, in general, not remain confined to a single plane; these rays are known as *skew rays*. Rays which remain confined to a plane (containing the axis) are known as *meridional rays*; obviously, all rays emanating from a point on the axis are *meridional rays*.

4.3 REFLECTION BY A SINGLE SPHERICAL SURFACE

We next consider the imaging of a point object O by a spherical mirror SPM (see Fig. 4.3) in the paraxial approximation; the point C represents the center of curvature. We proceed in a manner exactly similar to that in Sec. 4.2 and we refer to Fig. 4.3 to obtain

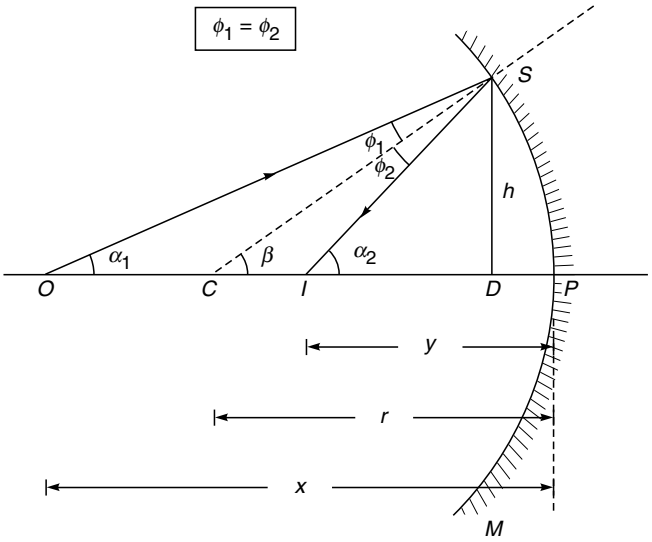


Fig. 4.3 Paraxial image formation by a spherical reflecting surface SPM.

$$\phi_1 = \beta - \alpha_1 \approx \frac{h}{r} - \frac{h}{x}$$

and

$$\phi_2 = \alpha_2 - \beta \approx \frac{h}{y} - \frac{h}{r}$$

where the distances x, y, h and r are defined in Fig. 4.3. Since $\phi_1 = \phi_2$ (the law of reflection), we get

$$\frac{1}{x} + \frac{1}{y} = \frac{2}{r} \qquad \textbf{(4.6)}$$

If we again use the sign convention that all the distances to the right of P are positive and those to its left negative, then $u = -x$, $v = -y$ and $R = -r$; thus we obtain the mirror equation

$$\frac{1}{u} + \frac{1}{v} = \frac{2}{R} \qquad \textbf{(4.7)}$$

which is the same as was derived by using Fermat's principle (see Problem 3.1). It is interesting to note that if we set $n_2 = -n_1$ in Eq. (4.5), we get Eq. (4.7). This follows from the fact that Snell's law of refraction Eq. (4.1) becomes the law of reflection if we have $n_2 = -n_1$.

We illustrate the use of Eq. (4.7) through an example.

Example 4.2 Consider an optical system consisting of a concave mirror $S_1P_1M_1$ and convex mirror $S_2P_2M_2$ of radii of curvatures 60 cm and 20 cm, respectively (see Fig. 4.4). We would like to determine the final image position of the object point O which is at a distance of 80 cm from the point P_1, the two mirrors being separated by a distance of 40 cm.

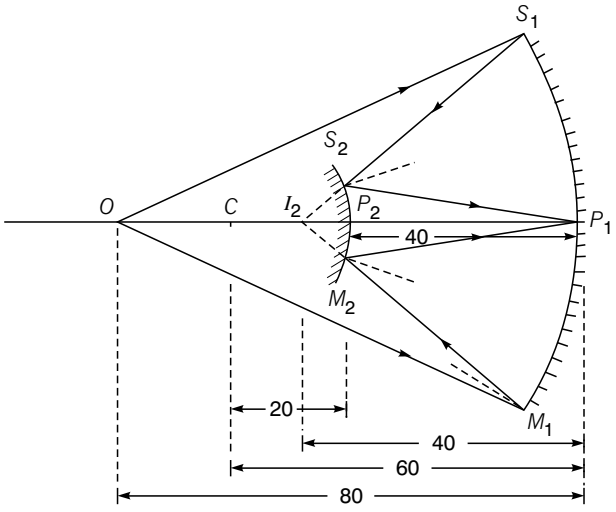


Fig. 4.4 Paraxial image formation by an optical system comprising a concave mirror $S_1P_1M_1$ and a convex mirror $S_2P_2M_2$.

We first consider the imaging by $S_1P_1M_1$; since $u = -80$ cm and $R = -60$ cm (because both O and C are on the left of P_1), we have

$$-\frac{1}{80} + \frac{1}{v} = -\frac{2}{60} \quad \Rightarrow \quad v = -48 \text{ cm}$$

In the absence of the mirror $S_2P_2M_2$, a real image will be formed at I_1 which now acts as a virtual object for $S_2P_2M_2$. Since I_1 is to the left of P_2, we have (considering imaging by $S_2P_2M_2$), $u = -8$ cm and $R = -20$ cm, giving

$$\frac{1}{v} - \frac{1}{8} = -\frac{2}{20} \quad \Rightarrow \quad v = +40 \text{ cm}$$

Thus, the final image is formed on the right of $S_2P_2M_2$ at a distance of 40 cm, which happens to be the point P_1.